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Inventor(s)	Watabe; Yusuke et al.

Compressor

Abstract

A compressor includes: a housing; a compressor body to compress gas; a motor to drive the compressor body; a control board to control the motor; and a tank to store the gas outputted from the compressor body, wherein the housing has: a machine chamber storing the compressor body and the motor; a B-chamber having piping to connect the compressor body with the tank; an A-chamber storing the tank; an A-partition separating the machine chamber from the A-chamber and B-chamber; and a B-partition between the A-chamber and the B-chamber, the A-chamber has an air intake port to introduce ambient air, the A-partition has an A-vent as a hole to communicate between the machine chamber and the A-chamber, and a suction port of the compressor body is located in vicinity to the A-vent.

Inventors:	Watabe; Yusuke (Tokyo, JP), Saito; Hiroaki (Tokyo, JP)
Applicant:	Hitachi Industrial Equipment Systems Co., Ltd. (Tokyo, JP)
Family ID:	83848274
Assignee:	HITACHI INDUSTRIAL EQUIPMENT SYSTEMS CO., LTD. (Tokyo, JP)
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Primary Examiner: Tremarche; Connor J

Attorney, Agent or Firm: Volpe Koenig

Background/Summary

TECHNICAL FIELD

(1) The present invention relates to a compressor.

BACKGROUND ART

(2) The background art in the present technical field includes those disclosed in Patent Literature 1 and Patent Literature 2. Patent Literature 1 discloses a compressor including, as a cooling means for an inverter (11) to drive an electric motor (9), a cooling air duct (12) dedicated to the inverter and a dedicated cooling fan (10) in the duct (12), to cool the inverter (11) by the cooling fan (10) and exhaust air after cooling to the outside, so that efficiency of cooling a compressor (8), the electric motor (9), and the like is improved.

(3) Patent Literature 2 is intended to provide a package compressor, including in a package a compressor body (2), a motor (3) to drive the compressor body (2), and an inverter (4) to control rotating the motor (3), wherein the package compressor is cooled such that restrictions on the arrangement of parts are reduced, while the inverter (4) is sufficiently cooled, to have improved productivity. The package compressor includes the compressor body (2) to compress air, the motor (3) to drive the compressor body (2), the inverter (4) to control a rotation speed of the motor (3), and a cooling fan (5) provided in the compressor body (2), wherein the inverter (4) is disposed in a path for taking in cooling air by the cooling fan (5) in the compressor body (2).

PRIOR ART DOCUMENTS

Patent Literatures

(4) Patent Literature 1: Japanese Patent Application Publication No. 2004-324615 (such as FIGS. 1, 2, and 3, and paragraphs 0025 and 0079), and Patent Literature 2: Japanese Patent Application Publication No. 2016-75159 (such as FIGS. 2, 3, and 5, and paragraphs 0010 and 0012).

SUMMARY OF THE INVENTION

Problems to be Solved

(5) However, Patent Literatures 1 and 2 each fail to describe a technique to reduce both noise and temperature of intake air in a package compressor. The present invention has been devised in view of the above-described situation and is intended to provide a compressor to reduce both noise and temperature of intake air.

Solution to Problem

(6) A compressor of the present invention, to solve the above-identified problems, includes: a housing forming a profile; a compressor body to compress gas; a motor to drive the compressor body; a control board to control the motor; and a tank to store the gas outputted from the compressor body, wherein the housing has: a machine chamber storing the compressor body and the motor; a B-chamber having piping to connect the compressor body with the tank; an A-chamber storing the tank; an A-partition separating the machine chamber from the A-chamber and B-chamber; and a B-partition between the A-chamber and the B-chamber, the A-chamber has an air intake port to introduce ambient air, the A-partition has an A-vent as a hole to communicate between the machine chamber and the A-chamber, and a suction port of the compressor body is located in vicinity to the A-vent.

Advantageous Effects of the Invention

(7) The present invention provides a compressor to reduce both noise and temperature of intake air.

Description

BRIEF DESCRIPTION OF DRAWINGS

(1) FIG. 1A is a schematic diagram of a scroll compressor according to a first embodiment, as viewed diagonally from above and front;

(2) FIG. 1B is a schematic diagram of the scroll compressor according to the first embodiment, with outer plates and a front A-partition of a housing thereof detached;

- (3) FIG. 2A is a schematic diagram of the scroll compressor according to the first embodiment, with the outer plates of the housing partially detached;
- (4) FIG. 2B is a schematic diagram of the scroll compressor, with the outer plates of the housing partially detached, as viewed obliquely from above, left, and rear;
- (5) FIG. 2C is a schematic diagram of the scroll compressor, with a rear plate of the housing detached, as viewed obliquely from above, left, and rear;
- (6) FIG. 3 is a schematic diagram of the scroll compressor according to the first embodiment, with the outer plates and an air tank detached, as viewed from right;
- (7) FIG. 4A is a schematic diagram of the scroll compressor according to the first embodiment, with the outer plates of the housing partially detached;
- (8) FIG. 4B is a schematic diagram of the scroll compressor according to the first embodiment, with a left side plate of the housing partially detached and a front A-partition detached, as viewed from left;
- (9) FIG. 5 is a schematic layout of the scroll compressor according to a second embodiment, as viewed from above;
- (10) FIG. 6 is a schematic layout of a scroll compressor in another case 1 of the second embodiment, as viewed from above;
- (11) FIG. 7 is a schematic diagram of a scroll compressor in another case 2 of the second embodiment, as viewed from left so as to face an A-chamber and a B-chamber;
- (12) FIG. 8 is a schematic diagram of a slit of a first modification in the front A-partition and a suction port of the scroll compressor body, as viewed from left;
- (13) FIG. 9 is a schematic diagram of a slit of a second modification in the front A-partition and the suction port of the scroll compressor body, as viewed from left FIG. 10 is a schematic diagram of a slit of a third modification in the front A-partition and the suction port of the scroll compressor body, as viewed from left;
- (14) FIG. 11A is a schematic diagram of a slit of a fourth modification in the front A-partition and the suction port of the scroll compressor body, as viewed from left;
- (15) FIG. 11B is a schematic diagram of a slit of the fourth modification in the front A-partition and the suction port of the scroll compressor body, as viewed from left;
- (16) FIG. 12A is a schematic diagram of a slit of a fifth modification in the front A-partition and the suction port of the scroll compressor body, as viewed from left;
- (17) FIG. 12B is a schematic diagram of a slit of the fifth modification in the front A-partition and the suction port of the scroll compressor body, as viewed from left;
- (18) FIG. 13A is a diagram of a scroll compressor according to a sixth modification, with a first front A-partition and a second front A-partition detached, as viewed from left so as to face the A-chamber and B-chamber; and
- (19) FIG. 13B is a diagram of the scroll compressor according to the sixth modification, with the first front A-partition not detached and the second front A-partition detached, as viewed from left so as to face the A-chamber and B-chamber.

DETAILED DESCRIPTION OF THE EMBODIMENTS

(20) Hereinafter, a description is given in detail of a scroll air compressor as an example of a compressor according to the present invention, with reference to the drawings as required.

First Embodiment

(21) FIG. 1A is a schematic diagram of a scroll compressor **S1** according to a first embodiment, as viewed obliquely from above and front. FIG. 1B is a schematic diagram of the scroll compressor **S1** according to the first embodiment 1, with a side plate **1i1**, a front plate **101m**, and a front A-partition **202a** (see FIG. 2A) of a housing **101** detached.

(22) The scroll compressor **S1** according to the first embodiment has its contour defined by the housing **101**, as shown in FIG. 1A. The housing **101** has a front plate **101m**, a top plate **101t**, a rear plate **101u** (see FIG. 2B), side plates **1i1** and **1i2**, and a bottom plate **1010**. An electrical component

box **204b**, containing an electrical component **204d** (see FIG. 2C) for controlling the scroll compressor **S1**, is provided inside the side plate **1i1**. The side plate **1i1** is provided with an air intake port **101a**.

(23) The housing **101** in FIG. 1B is provided therein with a scroll compressor body **102**, a motor **103**, a control board **104**, an air tank **105**, an electric fan **106**, and an air dryer **107**. The scroll compressor body **102** compresses air supplied through a suction port **109**. The motor **103** drives the scroll compressor body **102**. The control board **104** controls operation of the scroll air compressor **S1**. The air tank **105** stores compressed air generated by scroll compressor body **102**.

(24) The electric fan **106** cools the air tank **105** storing high-temperature compressed air. The air dryer **107** dehumidifies the compressed air stored in the air tank **105**. The air dryer **107** is connected to the air tank **105** by a pipe **108** serving as a flow path for the compressed air. The scroll compressor body **102** has the suction port **109** for sucking air. The suction port **109** is provided, on a peripheral surface **109s** thereof, with a filter **109f** for removing dust in the air. Note that the present embodiment includes the two suction ports **109** and two filters **109f**, as an example, but the number of the suction ports **109** and filters **109f** is not limited to two. For example, the number of the suction ports **109** and filters **109f** may be one, or three or more.

(25) The scroll compressor body **102** is connected by a rubber hose **110** with the air tank **105** storing compressed air generated by the scroll compressor body **102**. The top plate **101t** of the housing **101** has an exhaust port **101e** for the air dryer **107** and an exhaust port **101s** for the scroll compressor body **102** opened therein. The exhaust port **101e** is used for the air that has cooled the air dryer **107** to be discharged therethrough. The exhaust port **101s** is used for the air that has cooled the scroll compressor body **102** to be discharged therethrough.

(26) FIG. 2A is a schematic diagram of the scroll compressor **S1** according to the first embodiment, with the front plate **101m** and side plate **1i1** of the housing **101** detached. FIG. 2B is a schematic diagram of the scroll compressor **S1**, with the front plate **101m** and side plate **1i1** of the housing **101** detached, as viewed diagonally from above, left, and rear.

(27) FIG. 2C is a schematic diagram of the scroll compressor **S1**, with the rear plate **101u** of the housing **101** detached, as viewed obliquely from above, left, and rear. Note that a lid of the electrical component box **204b** is detached in FIG. 2C.

(28) <Machine Chamber **201**>

(29) As shown in FIG. 2A, the interior of the housing **101** is partitioned into a machine chamber **201**, an A-chamber **203**, and a B-chamber **204**. The housing **101** is provided therein with a front A-partition **202a** and a rear A-partition **202b** (see FIG. 3) to separate the machine chamber **201** from the rest, namely, the A-chamber **203** on a rear side and the B-chamber **204** on a front side. The front A-partition **202a** and rear A-partition **202b** constitute an A-partition **202**. The rear A-partition **202b** is fixed to the housing **101**. Note that the front A-partition **202a** and rear A-partition **202b** may be collectively referred to as the A-partition **202**.

(30) The front A-partition **202a** as the A-partition **202** is detachable by a user. FIG. 3 is a schematic diagram of the scroll compressor **S1** according to the first embodiment, with the outer plates of the housing **101** and the air tank **105** (see FIG. 2A) detached, as viewed from right. The front A-partition **202a** is provided, at an upper front portion thereof, with a handle **402** to be gripped by the user.

(31) As shown in FIG. 1B, the housing **101** is provided therein with a B-partition **205** to divide a space other than the machine chamber **201** into the A-chamber **203** and the B-chamber **204**. The machine chamber **201** houses the scroll compressor body **102** and the motor **103**.

(32) <A-Chamber **203**>

(33) The A-chamber **203** is provided therein with the air tank **105**. The A-chamber **203** on the rear side is formed, at a lower rear portion therein, with an air intake port **206** to introduce ambient air. Having the air intake port **206** provided behind a lower portion of the air tank **105** allows the air tank **105** to be cooled with cool air that has flowed through the air intake port **206**. As shown in

FIG. 4A, the electric fan **106** generates airflow within the A-chamber **203**, as a tank chamber, to cause cool air to flow from bottom up. In addition, cool air tends to collect at low points so that having the air intake port **206** at a low point is more effective than the other way around.

(34) As shown in FIG. 2B, the rear plate **101u** of the housing **101** behind the electric fan **106** is provided, at an upper portion thereof, with an exhaust port **101f**. In addition, the rear plate **101u** of the housing **101** is provided with an air intake port **101h** for the scroll compressor body **102** and air intake ports **101d1** and **101d2** for the air dryer **107**. As a result, the scroll compressor body **102** is cooled by the air entering through the air intake port **101h** of the rear plate **101u**. The air dryer **107** is cooled by the air entering through the air intake ports **101d1** and **101d2** of the rear plate **101u**.

(35) <B-Chamber **204**>

(36) The B-chamber **204** on the front side in FIG. 1B is provided therein with the pipe **108** to connect the air dryer **107** with the air tank **105**, and the rubber hose **110** to connect the scroll compressor body **102** with the air tank **105**. As shown in FIG. 2C, the B-chamber **204** is provided therein with the electrical component box **204b** storing the electrical component **204d**, an electromagnetic valve (not shown) for draining, and the like. In this way, the B-chamber **204**, having a comparably large amount of unused space, can accommodate various components such as the pipe **108**, the rubber hose **110**, the electrical component box **204b**, and the electromagnetic valve for draining.

(37) As shown in FIG. 3, the front A-partition **202a** and rear A-partition **202b** of the A-partition **202** separate the machine chamber **201** from the A-chamber **203** on the rear side and the B-chamber **204** on the front side.

(38) <Exhaust Duct **111**>

(39) As shown in FIG. 2C, the rear A-partition **202b**, a side duct partition **111a**, an upper duct partition **111b**, the bottom plate **101o**, and the top plate **101t** define an exhaust duct **111**. The exhaust duct **111** is provided therein with an airflow guide plate **111o**. The airflow guide plate **111o** reduces the speed of the air flowed to the exhaust duct **111** to promote heat absorption. A description is given of airflow in the exhaust duct **111**.

(40) Cool ambient air is taken into the housing **101** through the air intake port **101h** of the rear plate **101u** in FIG. 2B (see a solid thick arrow **a21** in FIG. 2C). The air that has entered the housing **101** comes around toward front points in the housing **101**, cools the scroll compressor body **102** (see FIG. 1B) installed at a lower front point in the housing **101**, and then flows to lower rear points in the housing **101** (see an open broken-line arrow **a22** in FIG. 2C). The air that has entered the housing **101** at lower rear points flows upward because a sirocco fan **102c** for cooling the scroll compressor body **102** blows air (see an open arrow **a23** in FIG. 2C). The upward airflow hits the airflow guide plate **111o** to slow down but still flows upward (see an open arrow **a24** in FIG. 2C), and is then discharged outside through the exhaust port **101s** in the top plate **101t** (gray arrow **a25** in FIG. 2C). Additionally, the airflow guide plate **111o** is arranged such that the bottom plate **101o** of the housing **101** cannot be seen through the exhaust port **101s** when viewed from above, so that a path for the upward airflow **a23** is blocked to have effect of reducing noise. In other words, having a path for airflow means to have vibrations also transmitted through the air in the path. Then, avoiding a straight airflow path contributes to reducing noise.

(41) <Vent **301**>

(42) As shown in FIG. 3, the rear A-partition **202b** is provided with at least a vent **301** (A-vent) as a hole to communicate between the machine chamber **201** and the A-chamber **203**. The electric fan **106** (see FIG. 4A) generates airflow within the A-chamber **203**, as a tank chamber, to cause cool air to flow from bottom up. In addition, cool air tends to collect at low points so that having the vent **301** (A-vent) at a low point is more effective than the other way around. Note that the “low point” includes a point vertically lower than the center and an upstream point in the airflow. The vent **301** is located in vicinity to the suction port **109** (see FIG. 1B) of the scroll compressor body **102** installed in the machine chamber **201**. Cool air flows to the scroll compressor body **102** through the

vent **301**. This causes the scroll compressor body **102** to be effectively cooled, to improve efficiency and performance.

(43) <Vent **302**>

(44) The front A-partition **202a** has a vent **302** (B-vent) to communicate with the B-chamber **204**. The air inside the B-chamber **204** enters the machine chamber **201** through the vent **302** and cools the scroll compressor body **102**.

(45) As shown in FIG. 2A, the rubber hose **110** runs through the vent **302** to connect the scroll compressor body **102** with the air tank **105**. This allows the rubber hose **110** to be cooled by the air flowing through the vent **302**.

(46) <Slit **303**>

(47) The front A-partition **202a** has a slit **303** to communicate between the machine chamber **201** and the B-chamber **204**. The slit **303** is different from the vent **301** (A-vent) in the rear A-partition **202b** and the vent **302** (B-vent) in the front A-partition **202a**, which are shown in FIG. 3. The slit **303** is provided in vicinity to the filter **109f** over the suction port **109** of the scroll compressor body **102**. This allows for effectively supplying cool air through the slit **303** to the filter **109f** over the suction port **109**. That is, the scroll compressor body **109** takes in cool air (air before being warmed by a device such as the motor **103**) via the filter **109f** through the suction port **109**, to improve compression efficiency and compression performance.

(48) The slit **303** has a rectangular shape with a short vertical dimension and a long horizontal dimension. The two slits **303** are positioned above the respective suction ports **109** of the scroll compressor body **102**. This is for the purpose of letting the air in the B-chamber **204** flow through the slits **303** as a laminar airflow, to interrupt the air warmed by the motor **103** provided above the scroll compressor body **102**. Air in the B-chamber **204** is supplied to the suction ports **109** of the scroll compressor body **102** through the slits **303**. This allows for supplying cool air in the B-chamber **204** through the slits **303** to the suction ports **109**.

(49) As shown in FIG. 3, the two slits **303** are provided below a parallel portion **108h** of the pipe **108** connecting the air dryer **107** with the air tank **105** so as to be parallel to the parallel portion **108h**, and above the respective suction ports **109** of the scroll compressor body **102**. Note that the upper slit **303** is provided between the lower slit **303** and the motor **103**, as viewed laterally. Laminar airflow caused by the shape and arrangement of the slits **303** allows for separating the airflow from the motor **103** and the air from the B-chamber **204**. In this way, the slits **303** are provided close to the pipe **108** for the air dryer **107** as well as the motor **103**. This allows for cooling the piping **108** and the motor **103**.

(50) The noise generated in the machine chamber **201** from the scroll compressor body **102** and the like is transmitted through the front A-partition **202a** and rear A-partition **202b**, echoes on inner wall surfaces of the housing **101** in the A-chamber **203** or the B-chamber **204**, and then is emitted out of the housing **101**. This allows for reducing a noise level of the scroll compressor **S1**.

(51) FIG. 4A is a schematic diagram of the scroll compressor **S1** according to the first embodiment, with the side plate **1i1** of the housing **101** and a part of the front plate **101m** detached. FIG. 4B shows a schematic diagram of the scroll compressor **S1** according to the first embodiment, with the side plate **1i1** of the housing **101** detached and the front A-partition **202a** detached, as viewed from left. A part of ambient air taken in through the air intake port **206** in the rear plate **101u** flows through the A-chamber **203** and enters the machine chamber **201** through the vent **301** (A-vent) in the rear A-partition **202b**, as shown in FIG. 3. The airflow through the A-vent **301** branches and flows toward the motor **103** and to the suction ports **109**. In particular, a flow path through the A-vent **301** via the motor **103** to a cooling fan for the motor (electric fan **106**) is different from a flow path through the A-vent **301** to the suction port **109**. In this way, cool air is supplied through the A-vent **301** to the scroll compressor body **102**. That is, the suction ports **109** are arranged so as to be close to the air intake port **206** in terms of a positional relationship with the sirocco fan **103c**. This allows for sucking cool air (air before being warmed by a device such as the motor **103**) flowing

through the air intake port **206** via the suction ports **109**, to improve compression efficiency. (52) In addition, the vent **301** in the rear A-partition **202b** in FIG. 3 is arranged between the electric fan **106**, as the cooling fan for the motor **103** (see FIG. 1B), and the suction port **109**. These measures reduce suction resistance through the suction port **109**, to improve compression performance. That is, the suction port **109** is positioned to suck cool air (air before being warmed by a device such as the motor **103**) through the vent **301** in terms of a positional relationship with the electric fan **106**, to contribute to improving compression efficiency.

(53) As shown in FIG. 4A, ambient air taken in from around the air dryer **107** located at an upper front point in the housing **101** is supplied through a C-vent **401** to the B-chamber **204** (as indicated by an open arrow all in FIG. 4A). The air supplied through the C-vent **401** (as indicated by the open arrow all in FIG. 4A) flows along the pipe **108** so as to be supplied to the scroll compressor body **102** through the slits **303** and the vent **302**. The pipe **108** is cooled by the airflow indicated by the open arrow all in FIG. 4A, so that the temperature of the compressed air to be supplied to the air dryer **107** is reduced to lessen a load for the air dryer **107**.

(54) In addition, air is supplied to the suction ports **109** of the scroll compressor body **102** through the slits **303** in the front A-partition **202a** in FIG. 4A. This prevents warmed air in exchange for cooling the motor **103** inside the machine chamber **201** from being sucked, so that the temperature of the intake air is reduced to improve compression performance. The front A-partition **202a** of the first embodiment is provided with a handle **402**, as shown in FIG. 4B. At a periodic checkout, a user can grip the handle **402** to pull out the front A-partition **202a**. In this way, having the front A-partition **202** provided with the handle **402** improves handling of the front A-partition **202a** when pulled out.

(55) As shown in FIG. 2A, the front A-partition **202a** is fixed at a front portion thereof, which can be easily accessed, by bolts **b1**, fitting, or the like. For example, the front A-partition **202a** may be fixed, with the front portion fastened by predetermined number of bolts and a projection at a bottom of the front A-partition **202a** fitted into a cut in the bottom plate **1010**. Note that the front A-partition **202a** may be fixed without using the bolts **b1**. When the front A-partition **202a** is fixed, a top end and a back end thereof are pressed via an elastic body **202d** (see FIG. 4B) to ensure airtightness, prevent vibration, and improve maintainability. Note that an elastic body **101d** (see FIG. 3) to face the top end is mounted on the housing **101**.

(56) According to the above configuration, the front A-partition **202a** and rear A-partition **202b** are provided, as shown in FIG. 3, to separate the machine chamber **201** from the A-chamber **203** on the rear side and the B-chamber **204** on the front side, as shown in FIG. 2A, and the B-partition **205** is provided to divide a space other than the machine chamber **201** into the A-chamber **203** and the B-chamber **204**. Then, the air intake port **206**, the vent **301**, the vent **302**, and the like are provided, as shown in FIGS. 2A and 3. This implements a scroll compressor to reduce both noise and temperature of intake air.

(57) Note that the position, quantity, and area of the vent **301** (see FIG. 3), vent **302**, and slit **303** may be changed.

Second Embodiment

(58) FIG. 5 is a schematic layout of a scroll compressor **S2** according to a second embodiment, as viewed from above. The scroll compressor **S2** of the second embodiment has the A-chamber **203** provided on the front side and the B-chamber **204** provided on the rear side. Other configurations are the same as those of the first embodiment. This results in the B-chamber **204** having a larger internal volume provided on the rear side, to allow for reducing a dimension of the compressor in depth to have a compressor reduced in size.

(59) Alternatively, the B-chamber **204** having a larger unused internal volume may be provided on the rear side to enlarge the exhaust duct **111** in FIG. 2C. This results in improving cooling performance of the sirocco fan **102c** (see FIG. 1B) to improve performance of the scroll compressor body **102** and thus performance of the scroll compressor **S1**.

(60) As shown in FIG. 5, the A-chamber **203** and the B-chamber **204** are laterally next to the machine chamber **201** and the exhaust duct **111**. The housing **101** of the scroll compressor **S2** is provided, at the bottom in a rear surface thereof defining the B-chamber **204**, with the air intake port **206** (also see FIG. 2B). The A-chamber **203** accommodates the air tank **105**. A lateral surface of the housing **101** defining the A-chamber **203** is provided, at the bottom therein, with an air intake port **203k**, as an opening, for cooling the air tank **105** in the A-chamber **203**.

(61) The machine chamber **201** accommodates the scroll compressor body **102** on a lower side, and the motor **103** for driving on an upper side. Accordingly, operation noise is generated in the machine chamber **201**. Effect of preventing noise may be enhanced with a noise absorbing material attached at a position which is away from the machine chamber **201** and where a large surface is available. Then, a noise absorbing material **204v** is attached to an inner surface of the housing **101** laterally defining the B-chamber **204**. Additionally, a partition **204s** for noise insulation may be provided in order to further enhance effect of preventing noise. Alternatively, a noise absorbing material **204v0** may be attached to the partition **204s**.

(62) The above-described configuration gives the scroll compressor **S2** to enhance effect of preventing noise. Note that only the partition **204s** for noise insulation may be provided without the noise absorbing material **204v**, or only the partition **204s** and noise absorbing material **204v0** may be provided. In addition, the scroll compressor **S1** as with the first embodiment may have a positional relationship among the machine chamber **201**, the A-chamber **203**, and the B-chamber **204** changed as in other cases below.

Another Case 1

(63) FIG. 6 is a schematic layout of a scroll compressor **S21** in another case 1 of the second embodiment, as viewed from above. The scroll compressor **S21** of another case 1 has the same layout of the machine chamber **201**, the A-chamber **203**, and the B-chamber **204** as the scroll compressor **S2** of the second embodiment. However, partitions **204s1**, **204s2**, **204s3**, and **204s4** for noise insulation are arranged such that front ends thereof are alternately aligned. This enhances effect of preventing noise.

(64) Note that the partitions **204s1**, **204s2**, **204s3**, and **204s4** for noise insulation may have noise absorbing materials **204v1**, **204v2**, **204v3**, and **204v4** respectively attached thereto. This further enhances effect of preventing noise. In addition, the scroll compressor **S1** of the first embodiment may have a positional relationship between the scroll compressor body **102** and motor **103** in the machine chamber **201** changed as in another case 2 below.

Another Case 2

(65) FIG. 7 is a schematic diagram of a scroll compressor **S22** in another case 2 of the second embodiment, as viewed from left so as to face the A-chamber **203** and B-chamber **204**. The scroll compressor **S22** of another case 2 has the motor **103** on the lower side in the machine chamber **201**, and the scroll compressor body **102** on the upper side. In this case, the scroll compressor body **102** is on the upper side and thus the suction port **109** is also on the upper side. Then, the two slits **303** are provided slightly below the respective suction ports **109** so as to be closer to the motor **103** on the lower side. Laminar airflow to the suction ports **109**, through the slits **303** having a short vertical dimension and a long horizontal dimension, generates a so-called air curtain, to prevent the air warmed in exchange for cooling the motor **103** on the lower side from being sucked into the suction port **109**.

(66) That is, when the motor **103** is on the lower side and scroll compressor body **102** is on the upper side within the machine chamber **201**, having the slits **303** at positions closer to the motor, with respect to the respective suction ports **109**, allows for preventing warmed air, after cooling the motor **103**, from entering the suction port **109**.

FIRST TO FIFTH MODIFICATIONS

(67) Increasing an area of the vent **301** (see FIG. 3) or the slit **303** (see FIG. 2A) of the scroll compressor **S1** according to the first embodiment, for example, increases volume of air to be

supplied to the suction port **109** (see FIG. **1B**) of the scroll compressor body **102**, to improve compression performance. The same reference signs are assigned to the same components as those of the first embodiment, and they are not described. Modifications are shown in which the slit **303** for the suction port **109** of the scroll compressor body **102** of the first embodiment is modified. The suction port **109** of the scroll compressor body **102** in FIGS. **8** to **12B**, as described below, is provided on a peripheral surface **109s** thereof with the filter **109f**.

First Modification

(68) FIG. **8** is a schematic diagram of a slit **303a** of a first modification in the front A-partition **202a** and the suction port **109** of the scroll compressor body **102**, as viewed from left. The slit **303a** of the first modification may form a flow path, with two or more holes provided in vicinity to the filter **109f** on the peripheral surface **109s** of the suction port **109**. The slit **303a** including the two or more holes are formed to have a short vertical dimension and a long horizontal dimension as a whole.

Second Modification

(69) FIG. **9** is a schematic diagram of a slit **303b** of a second modification in the front A-partition **202a** and the suction port **109** of the scroll compressor body **102**, as viewed from left. The slit **303b** of the second modification is a vertically long hole in vicinity to the filter **109f** on the peripheral surface **109s** of the suction port **109**. That is, the slit **303b** is a hole with a short horizontal dimension and a long vertical dimension.

Third Modification

(70) FIG. **10** is a schematic diagram of a slit **303c** of a third modification in the front A-partition **202a** and the suction port **109** of the scroll compressor body **102**, as viewed from left. The slit **303c** of the third modification is a horizontally long hole in vicinity to the filter **109f** on the peripheral surface **109s** of the suction port **109**. That is, the slit **303c** is a hole with a long horizontal dimension and a short vertical dimension.

Fourth Modification

(71) FIGS. **11A** and **11B** are schematic diagrams of slits **303d1** and **303d2** of a fourth modification, respectively, in the front A-partition **202a** and the suction port **109** of the scroll compressor body **102**, as viewed from left.

(72) The slit **303d1** of the fourth modification in FIG. **11A** includes three round holes in vicinity to the filter **109f** on the peripheral surface **109s** of the suction port **109**. The slit **303d1** in a round shape is a slit with a long vertical dimension and a short horizontal dimension in all three round holes.

(73) The slit **303d2** of the fourth modification in FIG. **11B** is a hole, having curvature distribution to form a flat shape, in vicinity to the filter **109f** on the peripheral surface **109s** of the suction port **109**. As described with the fourth modification, the slit **303** need not be a rectangular hole.

Fifth Modification

(74) FIGS. **12A** and **12B** are schematic diagrams of slits **303e1** and **303e2** of a fifth modification, respectively, in the front A-partition **202a** and the suction port **109** of the scroll compressor body **102**, as viewed from left. The fourth modification has a modified positional relationship between the suction port **109** of the scroll compressor body **102** and the slit **303**.

(75) The slits **303e1** of the fifth modification in FIG. **12A** are formed as horizontally long slits at diagonal positions in vicinity to the filter **109f** on the peripheral surface **109s** of the suction port **109**. The slits **303e2** of the fifth modification in FIG. **12B** are formed as two oblong slits at positions on either one of right and left sides, and in vicinity to the filter **109f** on the peripheral surface **109s**, of the suction port **109**. The first to fifth modifications as described above also allow for sucking unwarmed air through the suction ports **109** of the scroll compressor body **102**, while interrupting the air warmed in exchange for cooling the motor **103**.

Sixth Modification

(76) FIG. **13A** is a diagram of the scroll compressor **S22** according to a sixth modification, with a

first front A-partition **202a1** and a second front A-partition **202a2** detached, as viewed from left so as to face the B-chamber **204** and A-chamber **203**. FIG. **13B** is a diagram of the scroll compressor **S22** according to the sixth modification, with the first front A-partition **202a1** not detached and the second front A-partition **202a2** detached, as viewed from left so as to face the B-chamber **204** and A-chamber **203**.

(77) The sixth modification is configured to have the front A-partition **202a**, as described in the embodiments, divided into two, the first front A-partition **202a1** and the second front A-partition **202a2**. The front A-partition **202a** of the sixth modification has the first front A-partition **202a1** for separation from an area in the machine chamber **201** to store the motor **103**, and the second front A-partition **202a2** for separation from an area in the machine chamber **201** to store the scroll compressor body **102**. The first front A-partition **202a1** and second front A-partition **202a2** are attachable and detachable. The first front A-partition **202a1** is provided with a handle **402a** to be gripped. The second front A-partition **202a2** is provided with a handle **402b** to be gripped.

(78) The user can detach and attach the first front A-partition **202a1** and second front A-partition **202a2** by gripping the handles **402a** and **402b**, respectively, as shown in FIG. **13A**. In addition, the user can also pull out only the second front A-partition **202a2** by gripping the handle **402b**, without pulling out the first front A-partition **202a1**, as shown in FIG. **13B**. The user can also detach and attach only the first front A-partition **202a1** by gripping the handle **402a**, even though not shown. The sixth modification improves handling of the scroll compressor **S22**.

Other Embodiments

(79) 1) In the embodiments, the A-partition is composed of the front A-partition **202a** and the rear A-partition **202b**, with the rear A-partition **202b** fixed to the housing **101** and the front A-partition **202a** provided with the handle **402** (see FIG. **3**) so as to be detached, but the entire A-partition may be configured to have the handle **402** (see FIG. **3**) so as to be detached. This allows the user to grip the handle **402** to detach and attach the entire A-partition, to improve maintainability and handling.

(80) 2) Hereinabove, the embodiments and modifications have been described, but the present invention is not limited thereto and includes various modifications. For example, the embodiments and modifications have been described in detail for the purpose of illustrating the present invention, and are not necessarily limited to those having all the described configurations. In addition, the configuration of an embodiment or a modification may partly be replaced with the configuration of another modification, or the configuration of an embodiment or a modification may be added with the configuration of another embodiment or modification. Further, the configuration of each embodiment or modification may partly be removed, or added or replaced with another configuration.

(81) For example, the compressor body can be replaced with that of a compressor other than the scroll compressor, such as a screw compressor and a reciprocating compressor. From another perspective, the entire package may be arranged in a space filled with a specific gas, to compress any gas other than air, such as hydrogen gas, nitrogen gas, or Freon gas.

LIST OF REFERENCE SIGNS

(82) **101**: housing, **102**: scroll compressor body (compressor body), **103**: motor, **104**: control board, **105**: air tank (tank), **106**: electric fan (cooling fan), **107**: air dryer, **108**: pipe, **109**: suction port, **109f**: filter, **110**: rubber hose, **201**: machine chamber, **202**: A-partition, **202a**: front A-partition (A-partition), **202b**: rear A-partition (A-partition), **203**: A-chamber, **204**: B-chamber, **204d**: electrical component, **205**: B-partition, **206**: air intake port, **301**: A-vent (hole to communicate), **302**: B-vent, **303**: slit, **401**: C-bent, **402**: handle, and **S1**; **S2**; **S21**: compressor (scroll compressor).

Claims

1. A compressor comprising: a housing forming a profile; a compressor body to compress gas; a motor to drive the compressor body; a control board to control the motor; and a tank to store the

gas outputted from the compressor body, wherein the housing has: a machine chamber storing the compressor body and the motor, a B-chamber having piping to connect the compressor body with the tank, an A-chamber storing the tank, an A-partition separating the machine chamber from the A-chamber and B-chamber, a B-partition between the A-chamber and the B-chamber, and an exhaust duct used to exhaust air taken into the housing through a first air intake port in a rear plate of the housing and having cooled the compressor body, wherein the A-chamber has a second air intake port to introduce ambient air, the second air intake port is at a position in the rear plate near a bottom plate of the housing, wherein a first flow path is created by moving a portion of the ambient air from the second air intake port, through the A-chamber, past the tank, and out of the housing, wherein the A-partition has an A-vent as a hole to communicate between the machine chamber and the A-chamber, wherein a suction port of the compressor body is located in vicinity to the A-vent, and wherein a portion of the A-partition is included in components defining the exhaust duct.

2. The compressor as claimed in claim 1, further comprising: a cooling fan provided in the rear plate and the A-chamber at a location near a top plate of the housing, and wherein a second flow path through the A-vent to the cooling fan for the motor is different from a third flow path through the A-vent to the suction port.

3. The compressor as claimed in claim 1, wherein the A-partition has a B-vent to communicate with the B-chamber, and a slit to communicate between the machine chamber and the B-chamber, and the slit is located in vicinity to a suction port of the compressor body.

4. The compressor as claimed in claim 1, wherein a front A-partition, included in the A-partition, has a B-vent to communication with the B-chamber, and a slit to communicate between the machine chamber and the B-chamber, the slit is located in vicinity to the suction port, and the slit is at a point where air flowing along the pipe can be sucked, and which is in vicinity to a filter over the suction port.

5. The compressor as claimed in claim 1, wherein a front A-partition, included in the A-partition, has a B-vent to communicate with the B-chamber, and a slit to communicate between the machine chamber and the B-chamber, the slit is located in vicinity to the suction port, and the slit is in a shape having a vertical dimension different from a horizontal dimension.

6. The compressor as claimed in claim 1, wherein a front A-partition, included in the A-partition, has a B-vent to communicate with the B-chamber, and a slit to communicate between the machine chamber and the B-chamber, the slit is located in vicinity to the suction port, and the slit includes two or more members for the suction port.

7. The compressor as claimed in claim 1, wherein a front A-partition, included in the A-partition, has a B-vent to communicate with the B-chamber, and a slit to communicate between the machine chamber and the B-chamber, the slit is located in vicinity to the suction port, and the slit is between the suction port and the motor, as viewed laterally.

8. The compressor as claimed in claim 1, wherein the B-chamber is provided therein with a rubber hose to connect the scroll compressor body with the tank, and an electrical component for controlling the compressor.

9. The compressor as claimed in claim 1, wherein the A-partition can partly or wholly be detached and a detachable portion thereof is provided with a handle.

10. The compressor as claimed in claim 2, wherein the first flow path is different from both the second flow path and the third flow path.
